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AES standard for digital audio - Audio-embedded metadata - Part 2: MPEG-1 Layer II or MPEG-2 LSF Layer II

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AES standard for digital audio - Audio-embedded metadata - Part 2: MPEG-1 Layer II or MPEG-2 LSF Layer II

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Abstract

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio decoded from MPEG-1 Layer II or MPEG-2 LSF Layer II encoding. The data may be used to aid a subsequent re-encoding process. A method of carrying this data is described in Part 1 of this Standard.

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Foreword

This foreword is not part of the *AES41-2-2012 AES standard for digital audio - Audio-embedded metadata - Part 2: MPEG-1 Layer II or MPEG-2 LSF Layer II*.

This document describes a set of data that may be conveyed according to the method described in Part 1 of this Standard. It is one part of a multi-part revision of AES41-2009.

Digital compression is used for a significant amount of program acquisition, for program production and storage, and for distribution and broadcasting.

Compression systems add impairments to sound. The audibility of these impairments depends on the degree of compression and the techniques used. It has been proved that re-encoding with minimal further impairment can be achieved by using information describing the previous encoding process. The information that is preserved varies according to the encoding scheme that has been used. For example, for transparent cascaded coding of MPEG-1 Layer II or MPEG-2 Layer II audio, the information required by the downstream encoder includes:

- . the positions of the frame boundaries;
- . the bit rate of the compressed bit stream before decoding;
- . the coding mode (monophonic, joint stereo, and so on);
- . the bit allocation for each sub-band within a frame;
- . the scale factors for each sub-band within each sub-block of a frame.

In order to help with the process of editing decoded audio, it is useful to send additional coding-decision data to describe any timing offset that may be introduced by editing only at points corresponding to block boundaries (for example, 24-ms boundaries for MPEG-1 Layer II at 48 kHz). These data are particularly relevant for editing sound associated with co-timed video. The bit rate of these additional data is less than 30 kbit/s. A cyclic-redundancy check word must be appended to frames of data to provide a means to detect errors in the data.

This data signal can be used by a downstream encoder to guide its coding decisions. For example, if the audio signal is changed in any way (as it would by mixing it with another signal) then it could be inappropriate for the downstream encoder to re-use any previous coding decisions. However, in such a case, when a suitable transport mechanism is used, the coding-decision data would be corrupted so the coder could detect that the data are no longer valid.

This specification describes in detail the content of the coding-decision information, such as MPEG-header information, bit allocations, and scale factors, plus its representation in terms of the number, order, and meaning of its bits. A mechanism by which these data can be conveyed in a pulse-code-modulation audio signal is also described. The mechanism modifies one of the least significant bits of the audio word to signal the data. The precise time relationship of the data to the audio is also defined.

The draft of this document was developed by a writing group whose primary author was Andrew Mason.

John Grant
Chair, working group SC-02-02
2012-03

Note on normative language

In AES standards documents, sentences containing the word “shall” are requirements for compliance with the document. Sentences containing the verb “should” are strong suggestions (recommendations). Sentences giving permission use the verb “may”. Sentences expressing a possibility use the verb “can”.

AES standard for digital audio - Audio-embedded metadata - Part 2: MPEG-1 Layer II or MPEG-2 LSF Layer II

0 Introduction

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio decoded from MPEG-1 Layer II or MPEG-2 LSF Layer II encoding. The data may be used to aid a subsequent re-encoding process. This part should be read in conjunction with part 1.

0.1 Rationale for part 2 of this standard

Digital compression techniques allow the use of lower bandwidth channels and less storage space or, conversely, allow faster transmission and the storage of more program material. Digital compression will be used for a significant amount of program acquisition, for program production and storage, and for distribution and broadcasting.

Compression systems add impairments to sound. The audibility of these impairments depends on the degree of compression and the techniques used.

It has been proved that re-encoding with minimal impairment can be achieved by using information describing the previous encoding process. The information that is preserved varies according to the encoding scheme that has been used, but might include framing and quantization resolution data.

0.2 Patents

The Audio Engineering Society draws attention to the fact that it is claimed that compliance with this AES standard may involve the use of an application for patents concerning "Lip sync" and "Lip-sync with sub-frame error feedback."

The AES holds no position concerning the evidence, validity and scope of this patent right.

The holder of this patent right has assured the AES that it is willing to negotiate licenses under reasonable and non-discriminatory terms and conditions with applicants throughout the world. In this respect, the statement of the holder of this patent right is archived with the AES.

Information may be obtained from

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Attention is drawn to the possibility that some of the elements of this AES standard may be the subject of patent rights other than those identified herein. AES shall not be held responsible for identifying any or all such patent rights.

1 Scope

This document describes a format for the data to be transmitted to identify MPEG-1 Layer II and MPEG-2 LSF Layer II audio bit-rate reduction. This part assumes that the transmission mechanism according to part 1 of this Standard is used.

2 Normative references

The following standards contain provisions that, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the indicated standards.

AES41-1-xxxx, *AES standard for digital audio - Audio-embedded metadata - Part 1: General*, Audio Engineering Society, New York, NY., US.

ISO/IEC 11172-3 (1993-08), *Information technology - Coding of moving pictures and associated audio for digital storage media at up to about 1,5 Mbit/s - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

ISO/IEC 13818-3 (1998-05), *Information technology - Generic coding of moving picture and associated audio information - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

3 Definitions, symbols, and abbreviations

3.1 Definitions

3.1.1

analysis filter bank

filter bank in the encoder that transforms a broadband pulse-code-modulation (PCM) audio signal into a set of subsampled sub-band samples

3.1.2

channel

means of carrying an audio signal (for example, a sequence of data forming part of a bit stream)

3.1.3

coded-audio bit stream

coded representation of an audio signal

3.1.4

compression

reduction in the number of bits used to represent an item of data

3.1.5

CRC

cyclic redundancy check

3.1.6

dual-channel mode

mode where two audio channels with independent program contents (for example, bilingual) are encoded within one bit stream

NOTE The coding process is the same as for the stereo mode.

3.1.7

emphasis

filtering applied to an audio signal before storage or transmission (for example, to improve the signal-to-noise ratio at high frequencies)

3.1.8

joint stereo coding

any method that exploits stereophonic irrelevance or stereophonic redundancy

3.1.9

joint stereo mode

mode of the audio coding algorithm using joint stereo coding

3.1.10

layer

one of the levels in the coding hierarchy of the audio system defined in ISO/IEC 11172-3

3.1.11

multichannel

combination of audio channels conveying the audio signals used to create a spatial sound field

3.1.12

nextbits function

`nextbits()` function that permits comparison of a bit string with the next bits to be decoded in the bit stream

3.1.13

scale factor

factor by which a set of values is scaled before quantization

3.1.14

scale-factor index

numerical code for a scale factor

3.1.15

stereo mode

mode where two audio channels which form a stereophonic pair (left and right) are encoded within one bit stream

NOTE The coding process is the same as for the dual-channel mode.

3.1.16

sub-band

subdivision of the audio frequency band

3.1.17

sub-band filter bank

set of band filters covering the entire audio frequency range

NOTE In ISO/IEC 11172-3, the sub-band filter bank is a polyphase filter bank.

3.1.18

sub-band samples

filtered samples created by the sub-band filter bank within the audio encoder, constituting a filtered and subsampled representation of the input audio stream

NOTE From 384 time-consecutive input audio samples, twelve time-consecutive sub-band samples are generated within each of the 32 sub-bands.

3.1.19

synthesis filter bank

filter bank in the decoder that reconstructs a PCM audio signal from sub-band samples

3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

3.2.1 Arithmetic operators

3.2.1.1

+

addition

3.2.1.2

++

increment

3.2.1.3

--

decrement

3.2.1.4

*

multiplication

3.2.1.5

%

modulus operator, defined only for positive numbers

3.2.2 Relational operators

3.2.2.1

<

less than

3.2.2.2

==

equal to

3.2.2.3

!=

not equal to

3.2.3 Bitwise operators

&

AND

3.2.4 Assignment

=

assignment operator

3.3 Abbreviations

The following mnemonics are defined to describe the different data types used in the coded bit stream.

3.3.1

bslbf

bit string, left bit first, where "left" is the order in which bit strings are written in this standard.

NOTE Bit strings are written as a string of logic 1's and logic 0's within single quote marks, for example, '1000 0001'. Blanks within a bit string are for ease of reading and have no significance.

3.3.2

simsbf

signed integer, most significant bit first

3.3.3

uimsbf

unsigned integer, most significant bit first

4 Information for transparency

In order to re-encode decoded audio with minimal impairment, it is necessary to re-encode using the same coding decisions that were used by the previous generation (upstream) encoder. For example, for near-transparent cascaded coding of MPEG-1 Layer II or MPEG-2 Layer II audio, the information needed by the downstream encoder includes:

- the positions of the frame boundaries;
- the bit rate of the compressed bit stream before decoding;
- the coding mode (monophonic, joint stereo, and so on);
- the bit allocation for each sub-band within a frame;
- the scale factors for each sub-band within each sub-block within a frame.

When editing decoded audio, improved quality can be obtained if additional data are sent to describe any timing offset that could have been introduced by editing only at points corresponding to block boundaries (for example, at 48 kHz, 24-ms boundaries for MPEG-1 Layer II or MPEG-2 Layer II). This is particularly relevant for editing sound that has a close temporal relationship to accompanying pictures when loss of lip-movement synchronization is a problem.

This data signal can be used by a downstream encoder to guide its coding decisions. If the audio signal is changed in any way (for example, by mixing with another signal), then, in general, it would not be appropriate for the downstream encoder to re-use any previous coding decisions. However, in this case, if a suitable transport mechanism is used, the coding decision data become corrupted and the coder is able to detect that the data are no longer valid.

In this standard, it is assumed that audio coder control data are added to the least significant bit of the PCM audio signal in systems that handle the PCM audio with one of three different widths of PCM sample, namely, 16-bit, 20-bit, and 24-bit samples. In order to reduce the possibility of audibility of the coder-control data, the data are signaled by adjusting the parity of each 16-bit (20-or 24-bit) word appropriately. A CRC word is appended to frames of data to provide a means to detect errors in the data.

If the audio signal is changed in any way (for example, by mixing with another signal), then it is not appropriate that the downstream encoder re-use any previous coding decisions. However, in this case, the coder control data becomes corrupted and the coder is able to detect that the data are no longer valid.

5 Recoding_data for MPEG-1, or MPEG-2 low sampling frequency

5.1 AES41 metadata type

The presence of this data in the stream shall be indicated by **type**=1₁₀, or **type**=2₁₀, as defined in Part 1 of this standard.

5.2 Syntax

The syntax of **data()** is specified in the box below in accordance with Part 1.

Syntax	No. of bits	Mnemonic
<pre> data() { if(type==MPEG-1 or -2 Layer II){ ID bitrate_index sampling_frequency mode mode_extension copyright original/copy emphasis for(sb=0; sb<30; sb++){ allocation[sb] } for(sb=0; sb<30; sb++){ for(scfblock=0; scfblock<3; scfblock++){ scalefactor[sb][scfblock] } } for(p=data_bits%8; p != 0; p--){ padding_bit } } </pre>	1 4 2 2 2 1 1 2 4 6 5	bslbf bslbf bslbf bslbf bslbf bslbf bslbf uimsbf uimsbf bslbf

5.3 Semantics

The data are for MPEG-1 Layer II or for MPEG-2 LSF Layer II, where:

ID	is the algorithm ID, according to ISO/IEC 11172-3, that is set to logic 1 to indicate ISO/IEC 11172-3, and set to logic 0 to indicate extension to lower sampling frequencies specified in ISO/IEC 13818-3;
bitrate_index	indicates the bit rate, according to ISO/IEC 11172-3;
sampling_frequency	indicates the sampling frequency, according to ISO/IEC 11172-3;
mode	indicates the mode (monophonic, stereo, joint stereo, and so on), according to ISO/IEC 11172-3;
mode_extension	is used in joint stereo mode, according to ISO/IEC 11172-3;
copyright	indicates copyright protection according to ISO/IEC 11172-3;
original/copy	indicates whether copy is original or not, according to ISO/IEC 11172-3;
emphasis	indicates the type of de-emphasis to use, according to ISO/IEC 11172-3;
allocation	is the bit allocation index into the quantization table which shall be sent as a 4-bit integer (not a 2-, 3-, or 4-bit integer as would be used in the MPEG bit stream), with unused bits set to 0 and bit allocations for sub-bands that were not transmitted in the original coded bit stream set to 0;

scalefactors	are factors represented by a 6-bit integer scale-factor index which should be transmitted in the audio coder control data, even if the original bit stream used scale-factor select information to reduce the number sent, while scale factors for sub-bands that were not transmitted in the original coded bit stream should be set to 62 ₁₀ ;
padding_bit	indicates bits that are added to bring the total number of bits in the data to a multiple of 8 to match the length field.

6 Sample alignment of audio coder control data with respect to decoded-audio signal

6.1 General

Frame boundary positions in the decoded PCM audio can be determined from the position of the first bit of the audio coder control data. The relationship between the boundary and the position of this first bit depends on the coding scheme that was last used.

6.2 MPEG-1 Layer II or MPEG-2 Layer II

Because of the length of the response of the sub-band synthesis filters, when positioning the first coder control bit for MPEG-1 Layer II or MPEG-2 Layer II, a given frame of compressed data will give non-zero signal contributions extending beyond 24 ms (at a sampling frequency of 48 kHz) when decoded. However, the audio coder control data relating to a particular compressed audio frame may be carried on the PCM samples that are predominantly reconstructed from that particular frame (for example, for editing).

Therefore, the coder control data for a particular audio frame shall be added to the decoded signal at the output of the synthesis filter bank after a delay corresponding to the delay through the filter bank. This delay has been measured to be 240 sample periods.

An example showing the relationship between coded data and decoded data for typical MPEG-1 Layer II or MPEG-2 Layer II codecs is included in annex B. A method of transporting LSF (MPEG-2) signals on an AES3 interface, together with the audio coder control data signal, is described in annex C.

The LSF samples produced by the decoder shall be present, unmodified, in every second AES3 frame, when transported on an AES3 interface. In this case, the position of the first coder control bit shall indicate an unmodified LSF sample.

Annex A (informative) Using the audio coder control data

A.1 Derivation of MPEG scale-factor select information in the encoder

Note that scale-factor select information (**scfsi**) is not sent in the audio coder control data bit stream. An audio encoder may infer this information from the scale factors sent. The encoder should be able to identify scale factors that were not transmitted in the original bit stream and could, in the case where scale factors were duplicated in the original bit stream, reduce the number sent even further.

There will be redundant data in most cases, since it is to be expected that scale-factor select information will usually result in some scale factors not needing to be sent. However, the worst case would be where no saving can be made and all three scale factors for each sub-band in a frame have to be sent. If the coder control data are to have a fixed length, for ease of processing, then this worst case defines the format for transmission of scale factors.

A.2 The introduced time-offset field

The operation of a non-linear editing system, such as can now be based on a typical desk-top computer, usually requires the positioning of edit points according to events in the audio signal which are unlikely to coincide with frame boundaries in the coded bit stream. The situation might also arise that audio recordings are being conformed to an edit decision list that corresponds to video edit points. Once again, the edit points are unlikely to fall exactly on coded-audio frame boundaries.

In these situations, it might be better to move the audio edit points onto coded frame boundaries, so that a continuous decode and re-encode operation can be avoided. A transition might require decoding and re-encoding of the audio, for example to do a cross-fade between two signals, but once the transition has been effected, the sequence of frames could be continued without disturbance.

Figure A.1 shows two possibilities for editing together two bit streams.

The top part shows the edit points located arbitrarily within the bit stream. In neither signal does the edit point lie on a coded bit-stream frame boundary. After the edit, the frames from the second signal have to be decoded and re-encoded, at least partially, to maintain the continuity of the frame sequence in the final bit stream.

The bottom part shows the case where the edit point has been moved so that it now lies on frame boundaries in both signals. After the transition period is over, the frames from the second signal do not need to be processed before being put into the edited bit stream.

The penalty incurred is that the edit point is not in the position in which it was originally intended to be. A time offset has been introduced, shown as Δt .

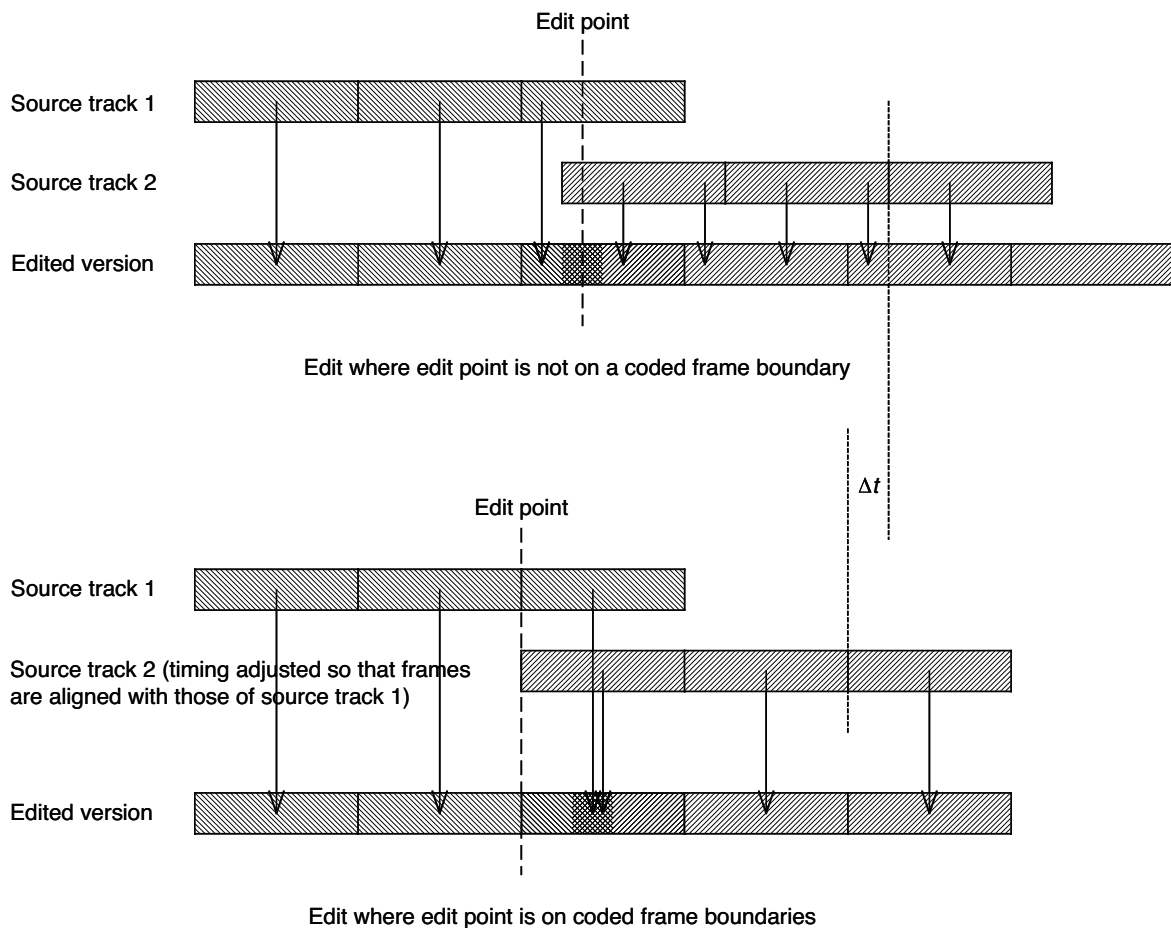


Figure A.1 - Alternative edit point locations relative to coded frame boundaries

Successive operations on the signals could result in several compromises on edit positions, each adding to the overall time offset. However, subsequent operations might make use of information about the size and sense (advance or delay) of this introduced time offset so as to minimize it.

A good example of how this might be used would be in multimedia work where the synchronization of sound and picture must stay within reasonable limits. When an operation is about to be performed on the audio signal that will affect the audio/video synchronization, timing adjustment can be made so that the cumulative effect is minimized.

A.3 Graphics of audio coder control data bit-stream syntax

This section shows graphically the bit-stream syntax for audio coder control data accompanying a signal decoded from a MPEG-1 Layer II or a MPEG-2 LSF Layer II bit stream. Fields are not fully described. Reserved fields are shown by shaded areas. Field lengths are indicated in bits. See Figure A.2.

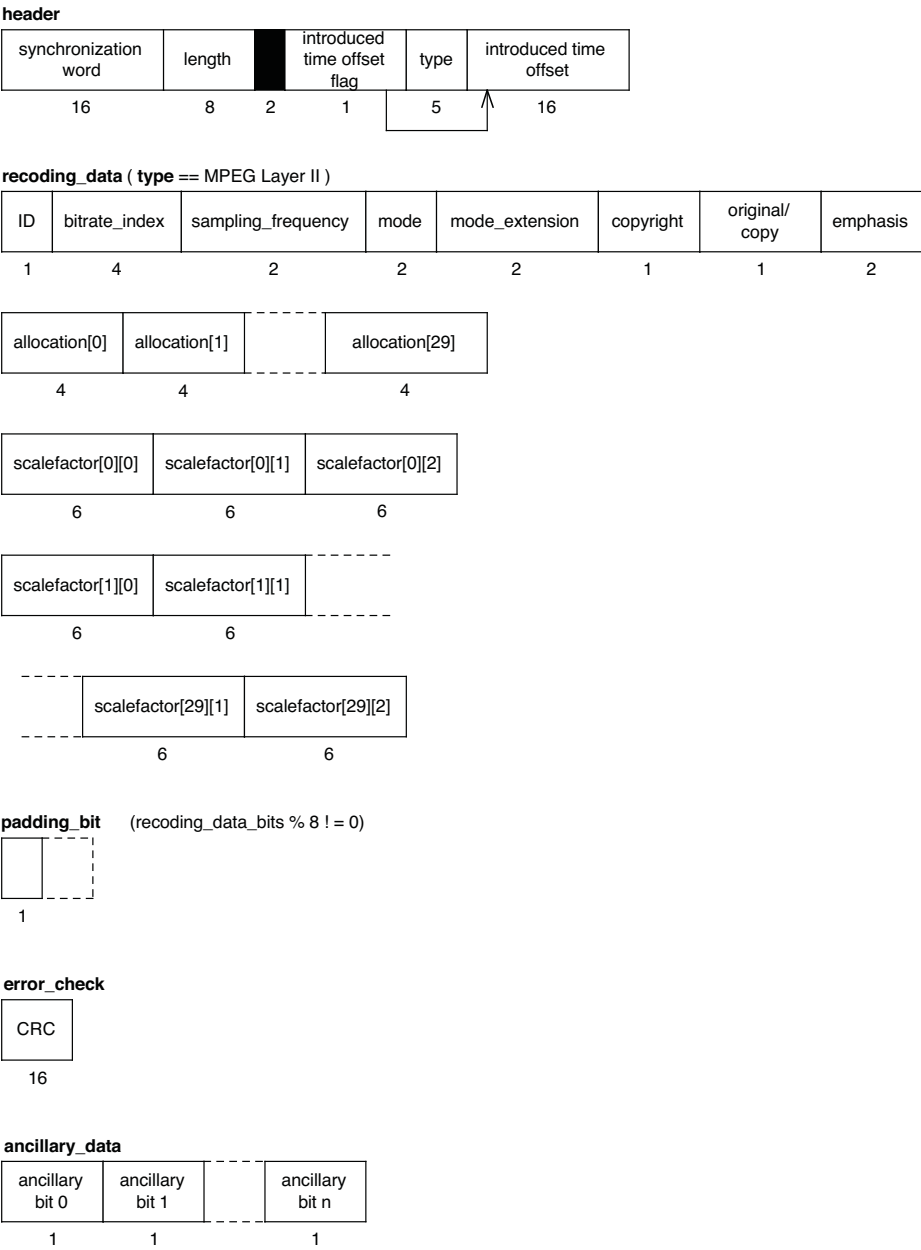


Figure A.2 - data_frame syntax diagram

Annex B (informative) Data alignment to decoded audio samples

When a PCM audio signal is fed through MPEG-1 Layer II or MPEG-2 Layer II analysis and synthesis filter, banks the signal that emerges is delayed by 481 samples. For the purposes of this discussion it will suffice to assume that the delay in the analysis filter bank is 241 samples and the delay in the synthesis filter bank is 240 samples.

The filter banks operate on blocks of 32 samples. In the following description, a sub-band sample will be referred to in the form $S[\text{block}][\text{subband}]$, where **subband** can take values from 0 to 31. A block of sub-band samples will be referred to as $S[\text{block}][0..31]$ and consists of one sample in each of 32 sub-bands.

A PCM sample will be referred to as $P[\text{block}][\text{sample number}]$. The sample number in the block takes values from 0 to 31.

B.1 Synthesis filter bank and audio coder control data generating decoder

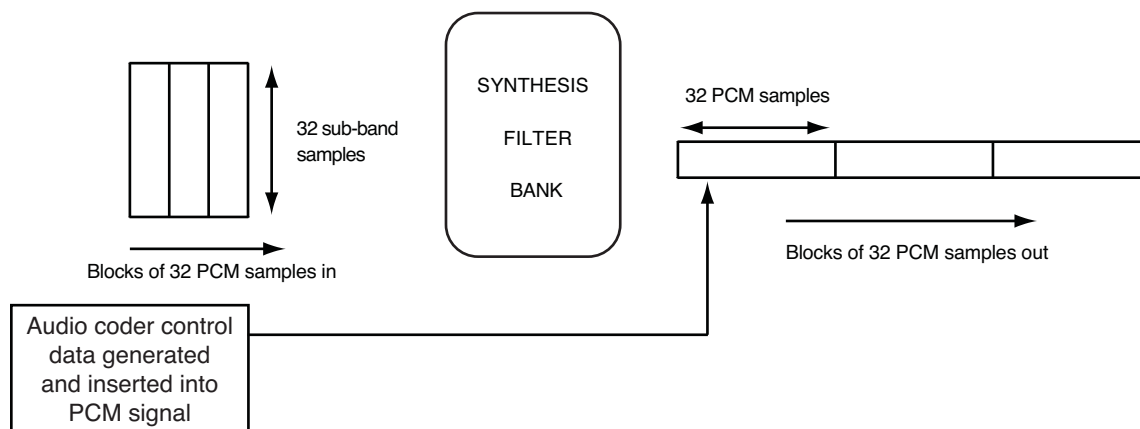


Figure B.1 — Operation of synthesis filter bank

Figure B.1 shows the operation of the synthesis filter bank in a decoder. A set of 32 sub-band samples (one sample in each of 32 sub-bands) enters the filter bank as a block, and a set of 32 time consecutive PCM samples emerges as a block.

When the block of sub-band samples $S[0][0..31]$ enters the filter bank, the block of PCM samples $P[0][0..31]$ emerges. Next, the block $S[1][0..31]$ enters and the block $P[1][0..31]$ emerges, and so on.

In an audio coder control data generating decoder, the coder control data should accompany PCM samples which are predominately related to the sub-band samples to which the coder control data refer. As discussed, the synthesis filter bank delay is taken to be 240 PCM sample periods, which is equal to 7,5 blocks of 32 samples. Therefore the audio coder control data describing the block of sub-band samples $S[0][0..31]$ should accompany the PCM samples $P[7][16..31]$ and $P[8][0..15]$.

The first bit of the audio coder control data frame synchronization word should be signaled 241 sample periods after the block of sub-band samples to which it relates are shifted into the synthesis filter bank in the decoder.

B.2 Analysis filter bank and audio coder control data assisted encoder

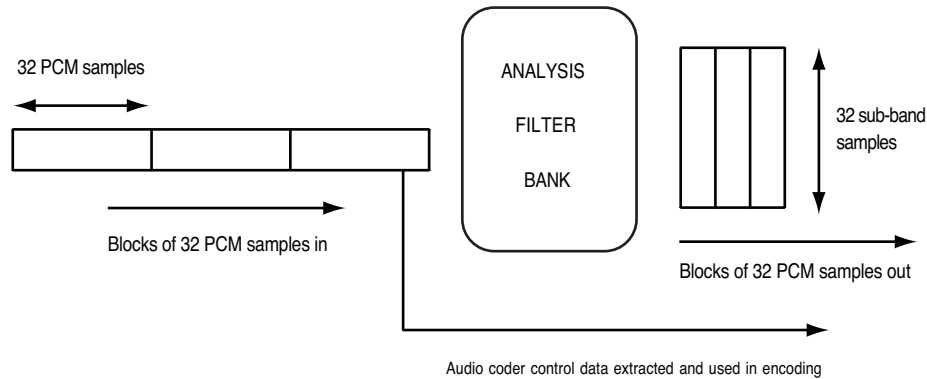


Figure B.2 - Operation of analysis filter bank

Figure B.2 shows the operation of the analysis filter bank in an encoder. A set of 32 PCM samples enters the filter bank as a block and a set of 32 sub-band samples, one in each of 32 sub-bands, emerges as a block.

Let us say that when the block of PCM samples $P[0][0..31]$ enters the filter bank the block of sub-band samples $S[0][0..31]$ emerges. For the audio coder control data to be used effectively, the blocks of 32 PCM samples that were used in the previous encoding should be fed into the filter bank in those same blocks, that is, the block boundaries should be in the same place. The audio coder control data can be used to establish this alignment.

The delay of the analysis filter bank is taken to be 241 PCM sample periods. Therefore, when the block of sub-band samples $S[n][0..31]$ emerges, the associated coder control data should have been carried with PCM samples $P[n-8][15..30]$ and $P[n-7][0..14]$.

Annex C (informative) Transmitting MPEG 2 LSF control data

The sampling frequencies defined for use with MPEG-2 LSF audio coding are not directly compatible with those supported by the AES3 interface. However, sampling frequencies twice the LSF frequencies are supported.

The practice described here is recommended for converting LSF signals into a form for transmission on the AES3 interface with audio coder control data.

The normal procedure for upconverting sampling frequencies in the simple ratio of 1 to 2 is to insert a zero-valued sample between each of the existing samples and then apply a low-pass reconstruction filter. The design of this reconstruction filter is crucial to the preservation of quality on re-encoding. It is recommended that a Nyquist filter be used. Implemented as a finite impulse response (FIR) transverse filter, this design would have every alternate coefficient zero with the exception of one. The filter would have an odd number of coefficients, and the central one would be the one alternate sample that was not zero. Figure C.1 shows a typical set of filter coefficients.

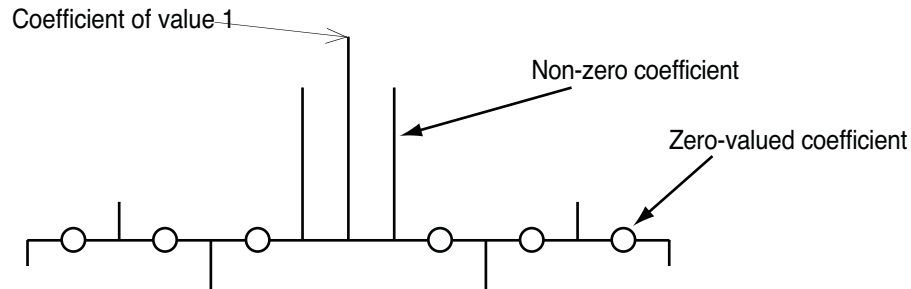


Figure C.1 — Example set of Nyquist filter coefficients

The benefit of using this filter in this application is that the original sample values in the LSF signal are preserved in the filtered signal. The inserted zero-valued samples are replaced by filtered versions. Provided that it is known which of the alternate values in the reconstructed signal is derived from the decoding of the coded bit stream and which is derived from filtering, a re-encoding process can be performed which will have as its input exactly the same sample values as were produced by the previous decoding operation. The effect of sample interpolation would have no effect on these sample values provided that the operation uses a Nyquist filter as described.

The audio coder control data signal should be used to indicate which of two alternate samples is derived from the decoding (a decoded sample) and which is produced by filtering (a filtered sample). The first bit of the audio coder control data synchronization word should be signaled in the filtered sample before the relevant decoded one defined by the sample alignment specification given in annex B.